

Basic Principles of Medicine 2
Module : Digestion and Metabolism
Lecture No: 21

Cholesterol Metabolism and Regulation

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Functions of Cholesterol

- Very Hydrophobic; Contains 4 ring sterol nucleus.
- In blood, found as cholesterol esters in lipoproteins (LDL and HDL)
- Functions of cholesterol
 - Precursor for **vitamin D**
 - Component of cell membrane – fluidity
 - Used for **bile acid synthesis** only in **liver**
 - Cholesterol synthesis occurs in the **brain** due to the blood-brain barrier
 - Precursor for **steroid hormones**
 - In **adrenal cortex** (cortisol, aldosterone, androgens)
 - In **ovaries** (estrogen and progesterone)
 - In **testes** (testosterone)

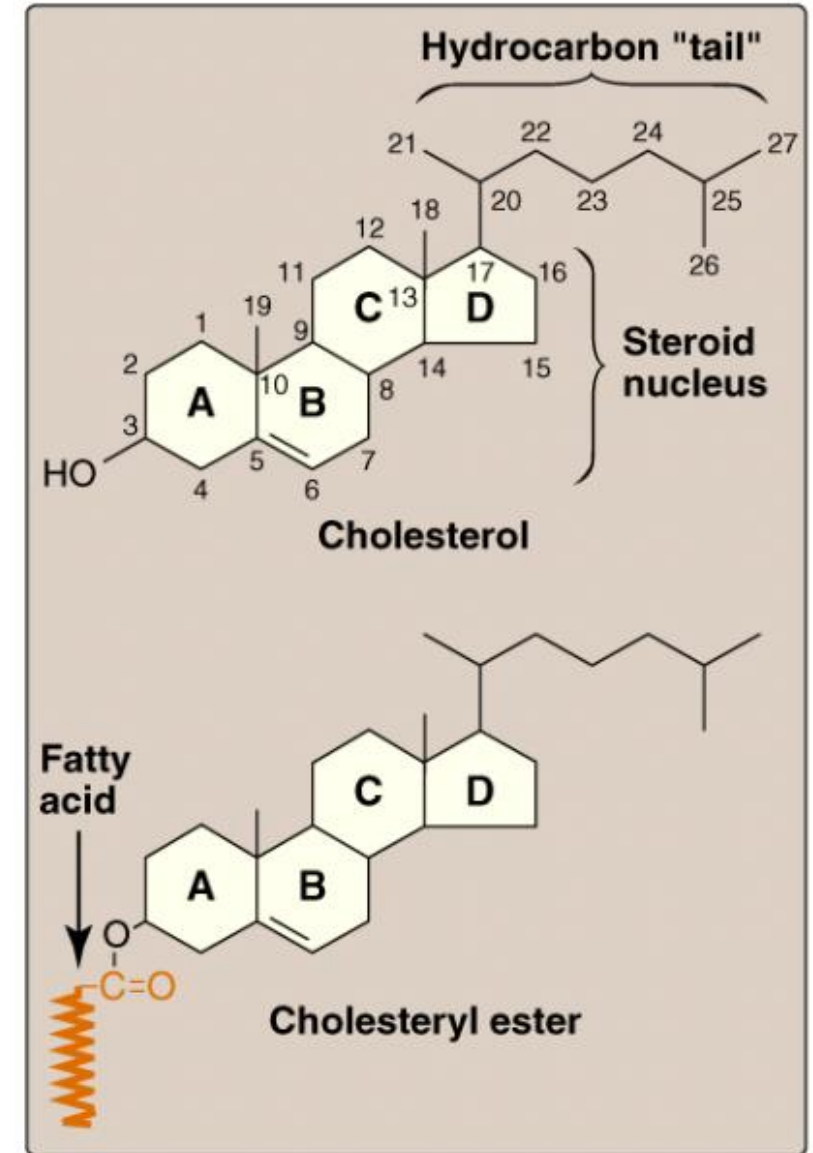


Figure 18.2

Recommended Reading

- **Lippincott's Biochemistry 9th Edition:**

 <https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402>

- **Chapter 18: Cholesterol, Lipoprotein, and Steroid Metabolism** [Cholesterol, Lipoprotein, and Steroid Metabolism | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)

- Question: 18.1
- Concept map: p.268

Objectives: Cholesterol Metabolism and regulation

SOM. MK.I.BPM2.3.DM.2.BCHM.1210	Outline cholesterol synthesis and the role of acetylCoA as a precursor.
SOM. MK.I.BPM2.3.DM.2.BCHM.1211	Indicate the biochemical defect in Smith-Lemli-Opitz syndrome (SLOS).
SOM. MK.I.BPM2.3.DM.2.BCHM.1212	Identify the sources of NADPH for cholesterol synthesis: pentose phosphate pathway.
SOM. MK.I.BPM2.3.DM.2.BCHM.1213	Explain regulation of cholesterol synthesis at the level of HMG-CoA reductase.
SOM. MK.I.BPM2.3.DM.2.BCHM.1214	Outline bile acid synthesis and its regulation and bile salt formation.
SOM. MK.I.BPM2.3.DM.2.BCHM.1215	Describe the role of statins in the management of hypercholesterolemia

Sources of Cholesterol

✓ **Diet: animal sources**

Cholesterol only
comes from
animal sources

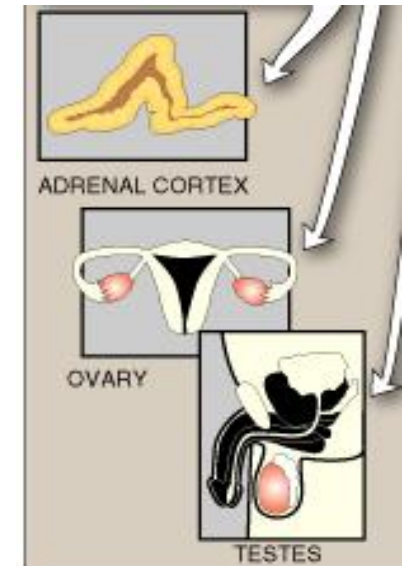
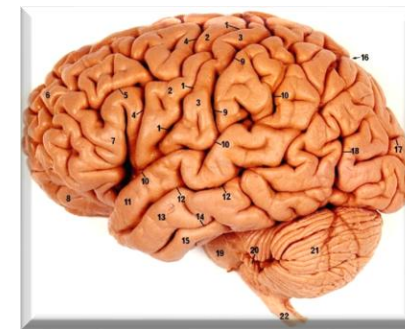
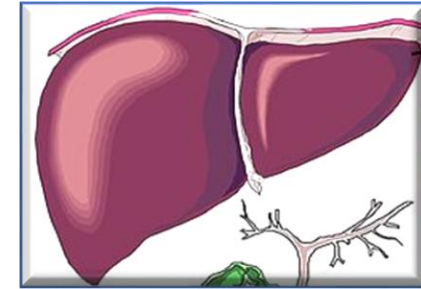


No plant food
contains cholesterol

✓ ***De novo* synthesis**

✓ **Sites of *De novo* synthesis of cholesterol**

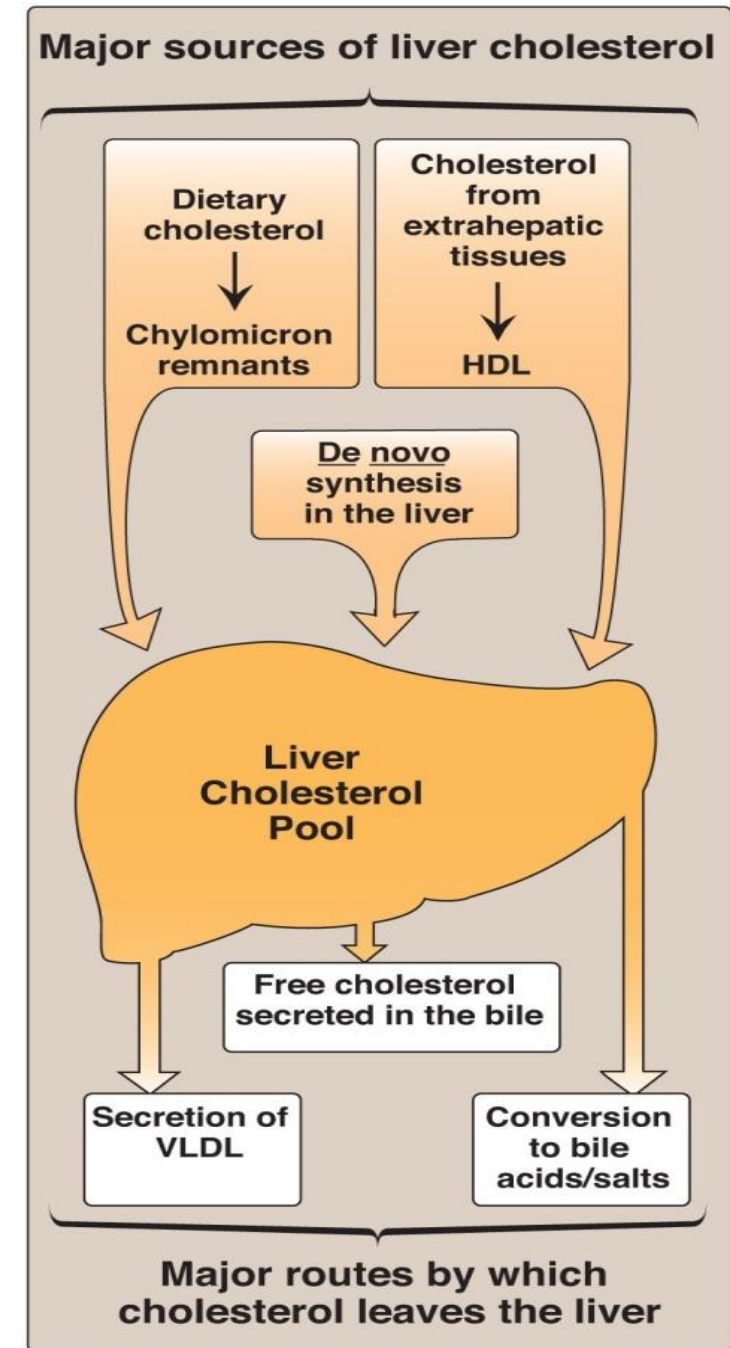
- Primarily in the **liver**.
- Brain
- Intestine
- Gonads and Reproductive tissues, including
 - Ovaries,
 - Testes,
 - Placenta
- Adrenal cortex



Cholesterol metabolism in liver

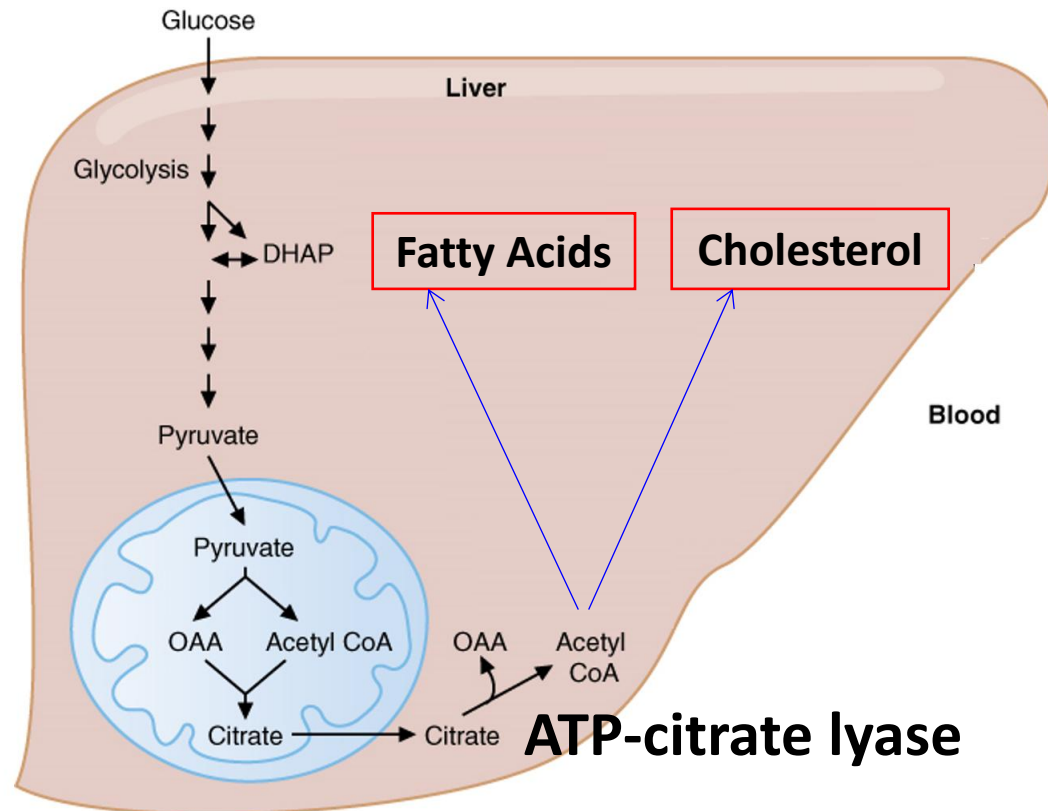
Cholesterol metabolism is an intricate balance:
biosynthesis/ transport/ utilization

Liver: Central role in cholesterol homeostasis.



Cholesterol Synthesis

- Begins in **cytosol**
- Acetyl-CoA is source of carbon units
- Mitochondrial acetyl-CoA formed from pyruvate (glucose) → Citrate
- Cytosolic **ATP-citrate lyase** forms acetyl CoA in cytosol



- In the **well-fed** state, in **hepatocytes**, **insulin** favors the pathways of:
 - Glycolysis
 - Glycogen synthesis
 - De novo synthesis of **Fatty acids**
 - **Cholesterol synthesis**

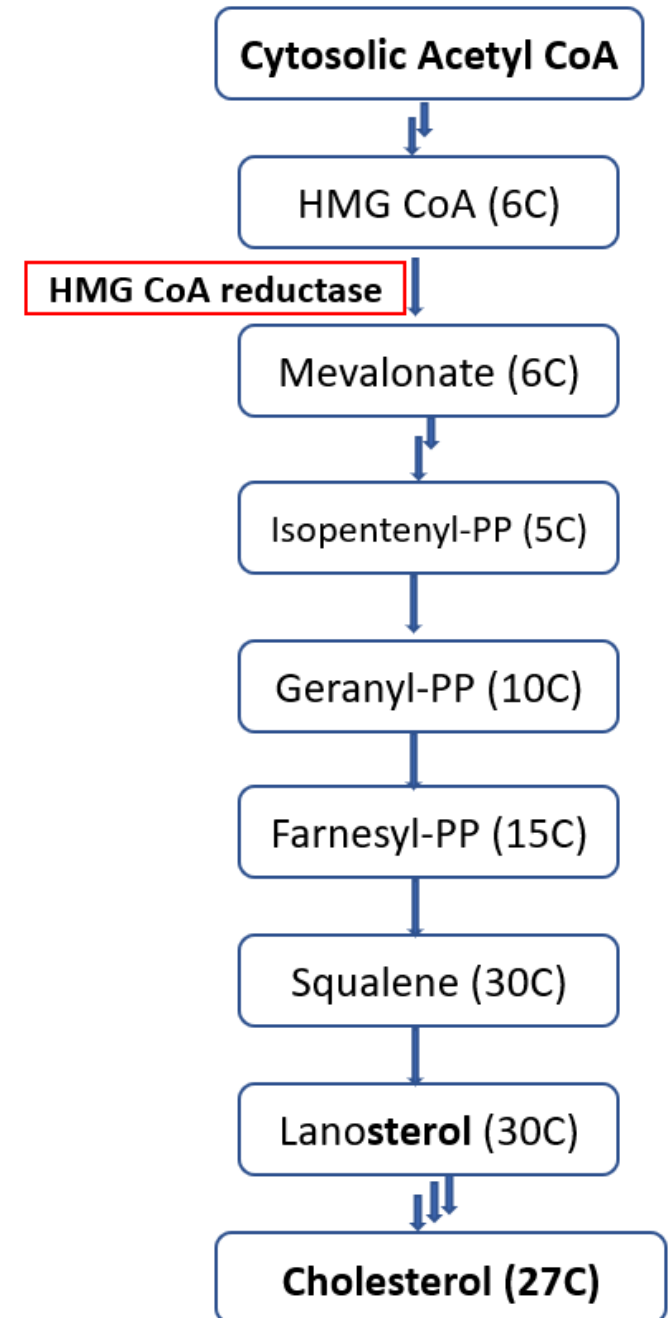
Sources of NADPH for Cholesterol Synthesis

- NADPH provides reducing equivalents for cholesterol biosynthesis
- NADPH is produced by:
 1. Pentose phosphate pathway
 - Major pathway
 2. Malic enzyme

Overview of Cholesterol Synthesis pathway

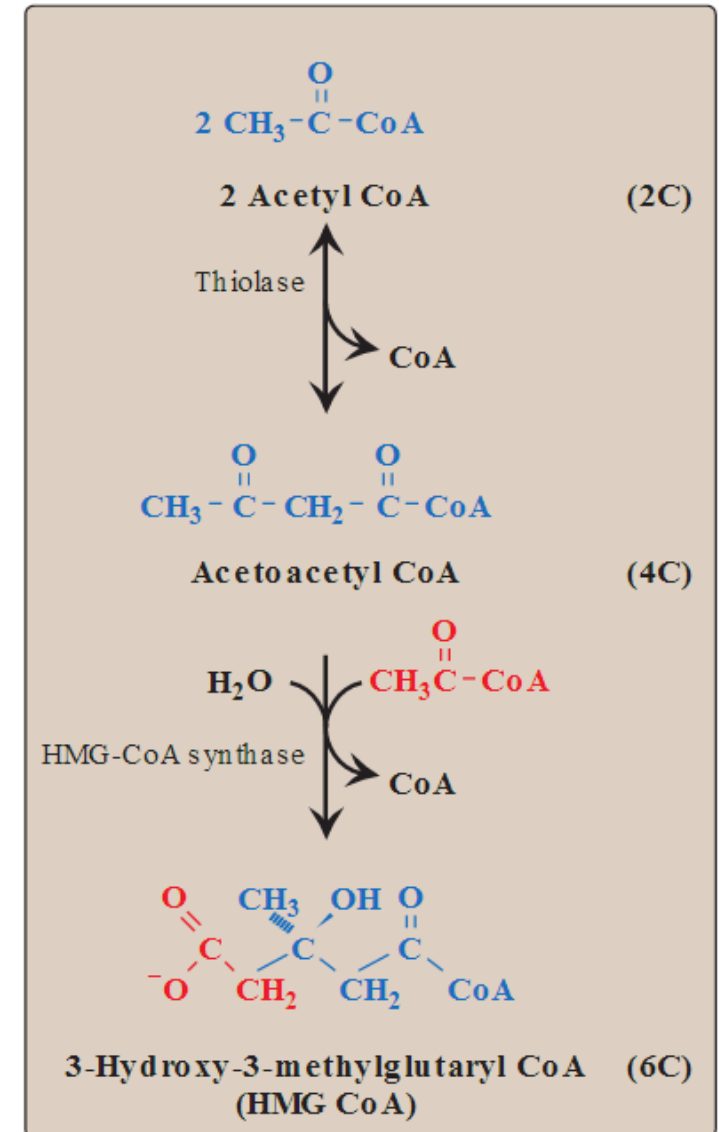
- ✓ Cytosolic **acetyl CoA** (formed by **ATP-citrate lyase** from citrate)
- ✓ **HMG CoA** is synthesized from acetyl CoAs in cytosol
- ✓ **Mevalonate** formed in the *regulated step* catalyzed by **HMG CoA reductase** (Uses **NADPH**)
- ✓ **Mevalonate-PP** → **Isopentenyl-PP (IPP)** (5-C building block)
- ✓ The 5-C IPP used to form **geranyl-PP** (10 C) and then **farnesyl-PP** (15 C).
- ✓ Two farnesyl-PP are linked to each other to form **squalene** (30 C).
- ✓ The **first sterol** formed is **lanosterol** (30 C)
- ✓ After many steps **cholesterol** (27 C) is formed. Uses **NADPH**

Note: Synthesis of cholesterol uses many molecules of **NADPH** (**HMG CoA reductase** and conversion of lanosterol to cholesterol)



Formation of 3-hydroxy-3-methylglutaryl coenzyme A (HMG –CoA)

- Involves 3 molecules of acetyl-CoA
- **Cytosolic HMG-CoA synthase** catalyses the linkage of 3rd Acetyl-CoA.
 - Resulting in formation of **HMG-CoA**
- Remember: **Mitochondrial** HMG-CoA synthase is regulated enzyme of ketogenesis



HMG-CoA reductase: Formation of Mevalonate

- HMG CoA is reduced to mevalonate
- Catalyzed by **HMG-CoA reductase** (Rate limiting enzyme)
- Requires **NADPH** (reducing equivalent)
- **Primary regulatory step of cholesterol biosynthesis**

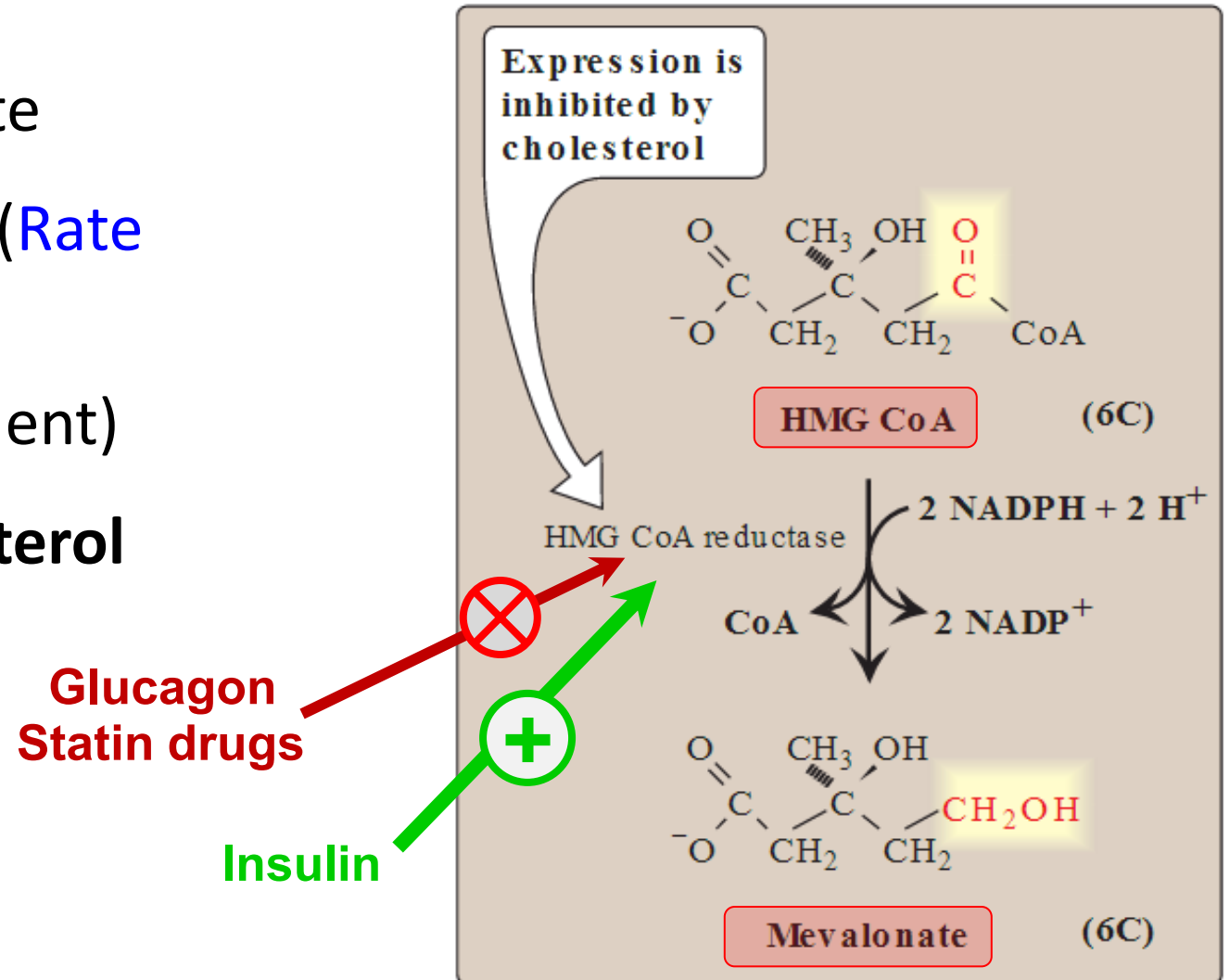
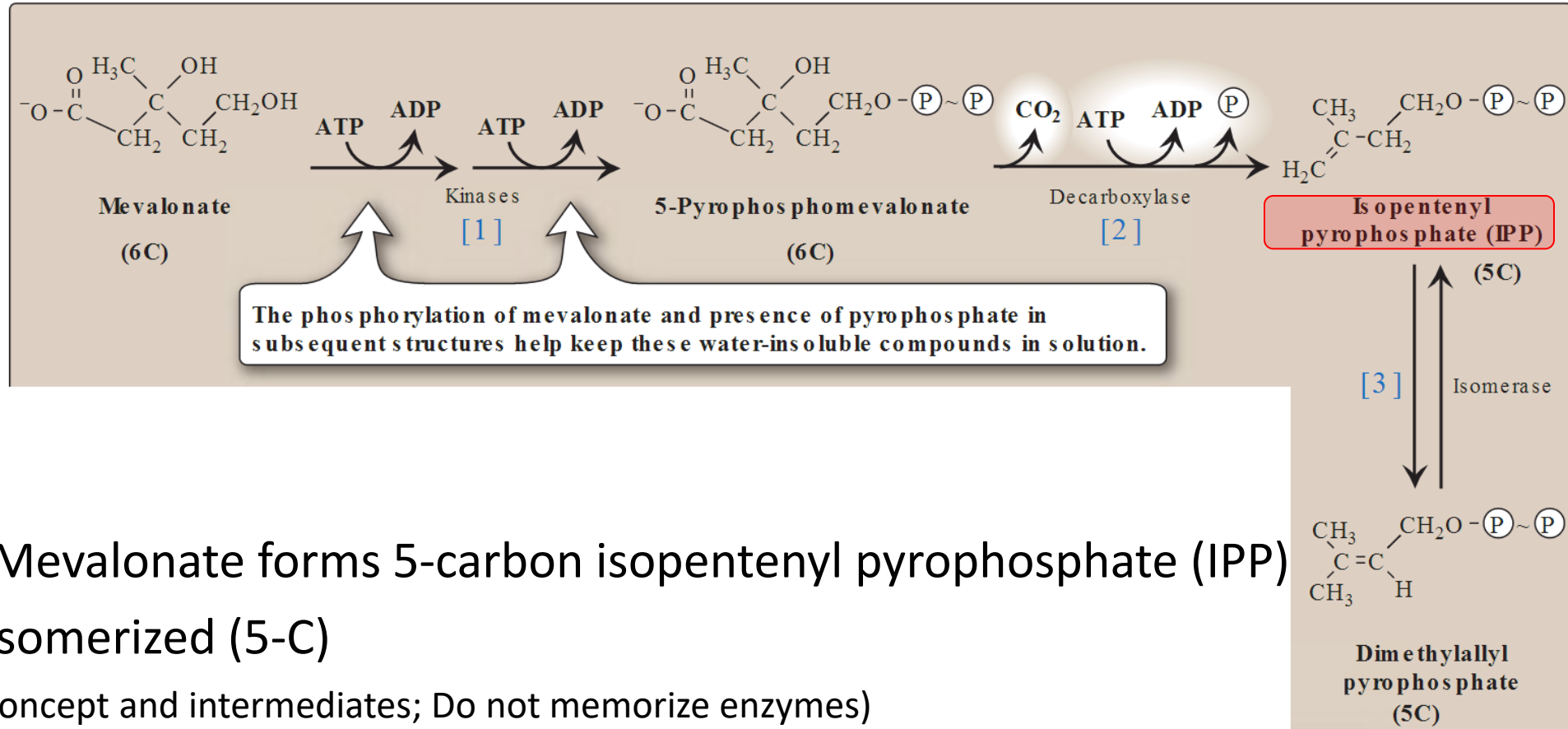


Figure 18.4

Synthesis of Cholesterol from Mevalonate

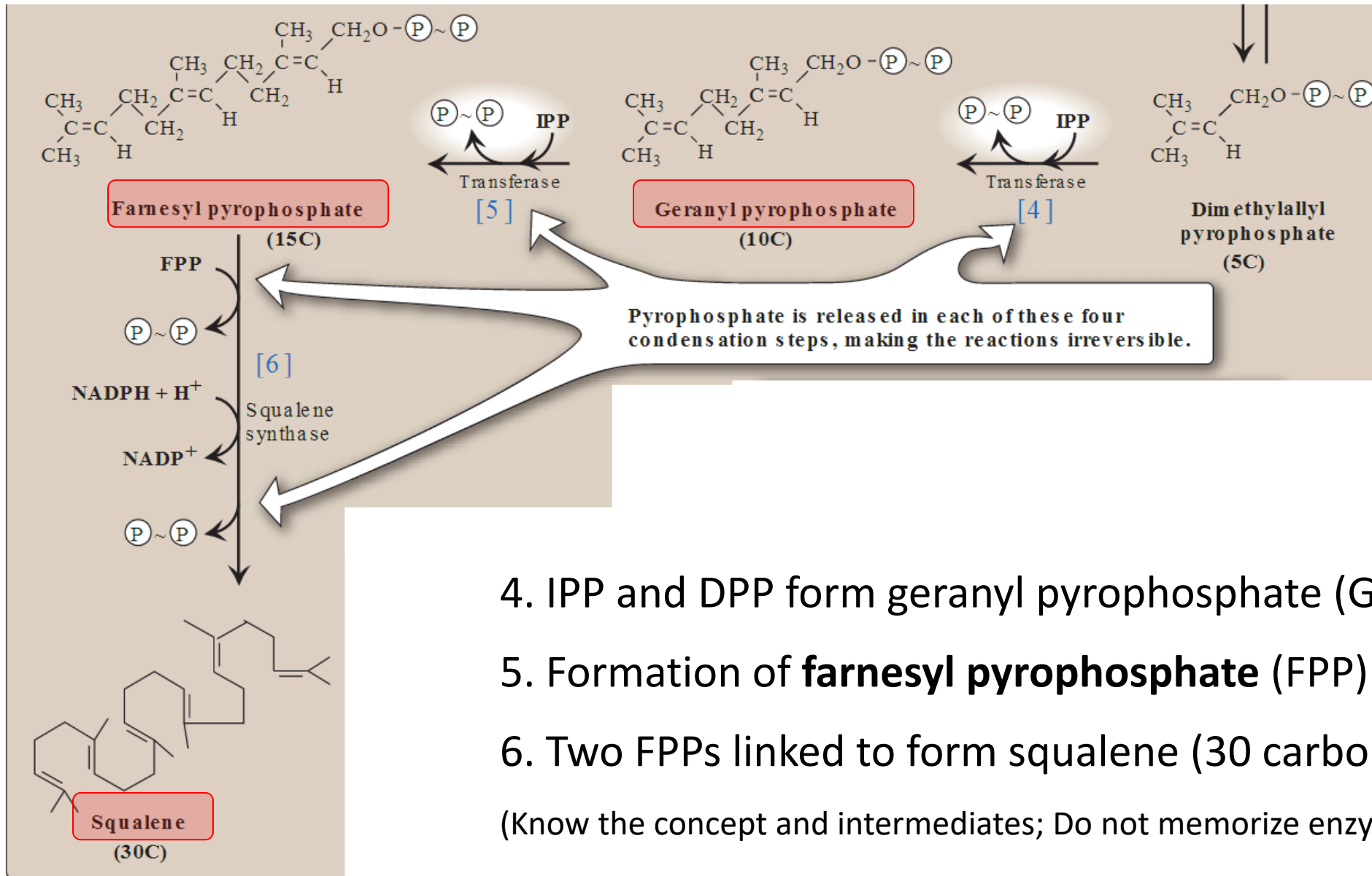


1 and 2. Mevalonate forms 5-carbon isopentenyl pyrophosphate (IPP)

3. IPP is isomerized (5-C)

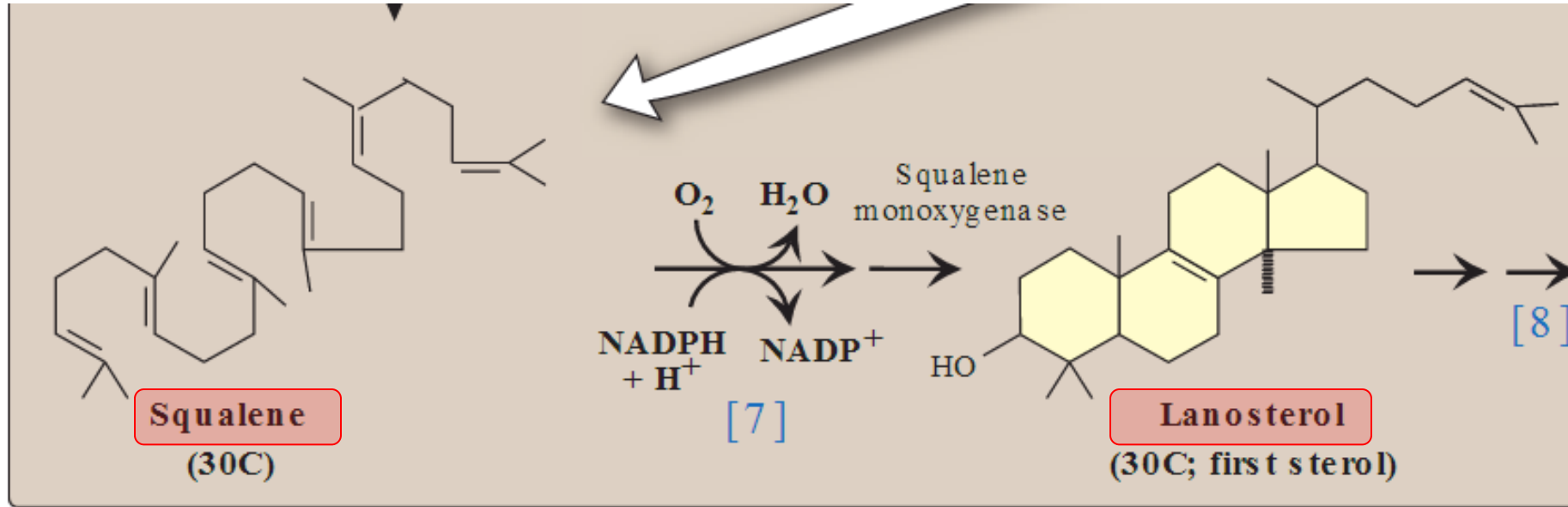
(Know the concept and intermediates; Do not memorize enzymes)

Synthesis of Cholesterol - continued



4. IPP and DPP form geranyl pyrophosphate (GPP) (10 C)
 5. Formation of **farnesyl pyrophosphate** (FPP) (15 C)
 6. Two FPPs linked to form squalene (30 carbons)
- (Know the concept and intermediates; Do not memorize enzymes)

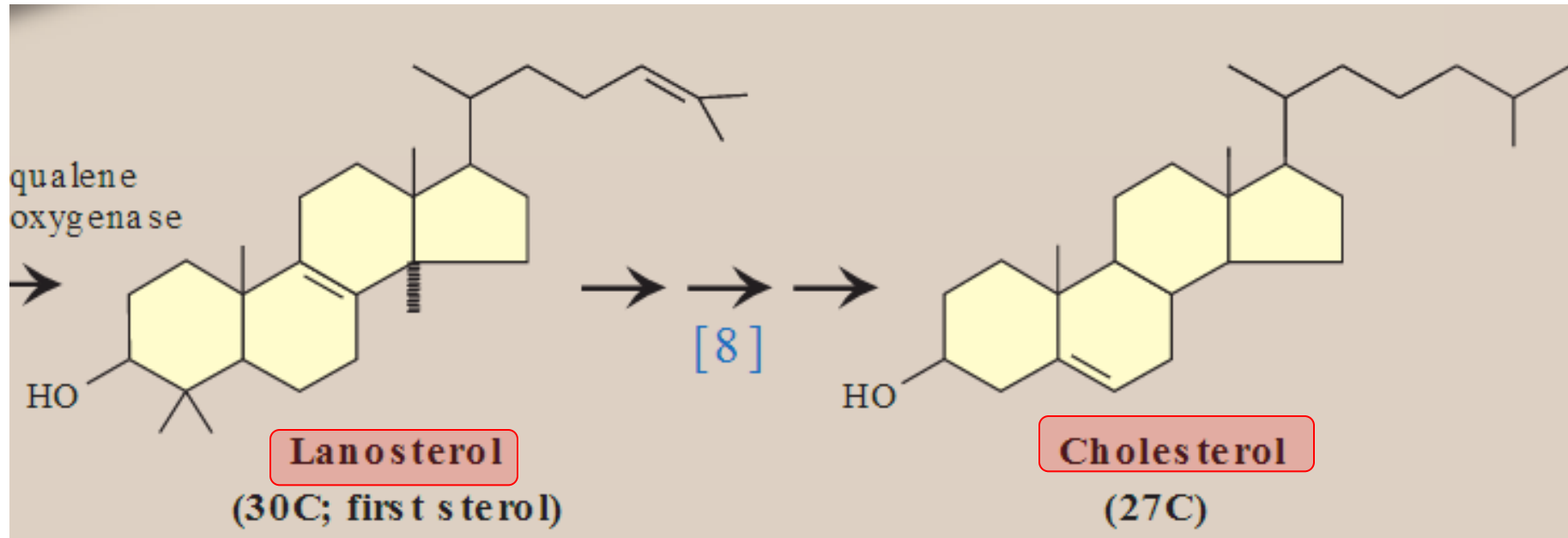
Synthesis of Cholesterol - continued



7. Squalene cyclized to form lanosterol (Uses NADPH)

- Lanosterol is the first sterol ring formed
- (Know the concept and intermediates; Do not memorize enzymes)

Synthesis of Cholesterol - continued

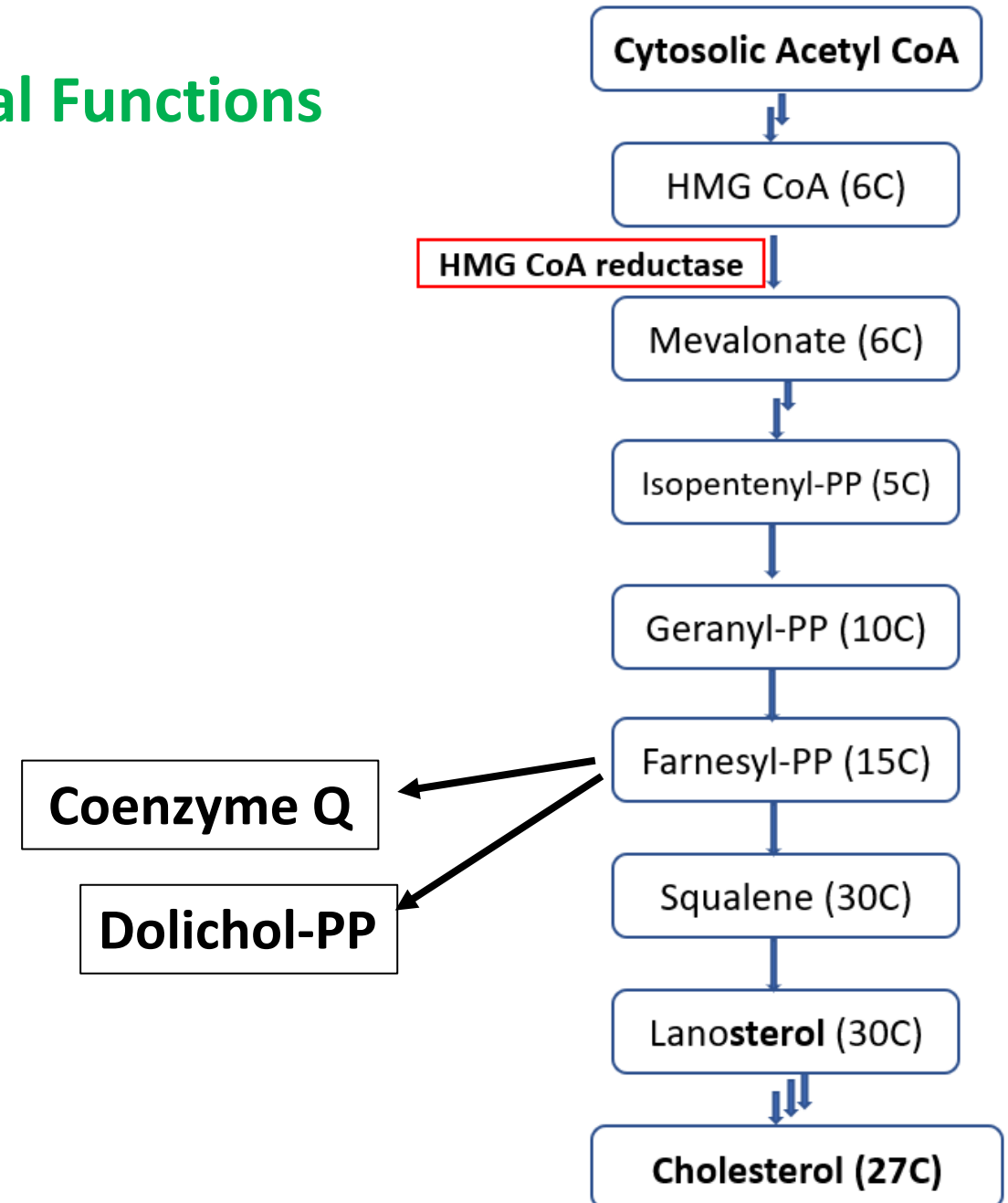


8. **Lanosterol (30C)** converted to **cholesterol (27C)** by a series of reactions in the ER by many enzymes

- Lanosterol converted to cholesterol via several steps
- Requires **NADPH** and removal of 3-carbon atoms
- **Finally**, a 27-carbon cholesterol formed

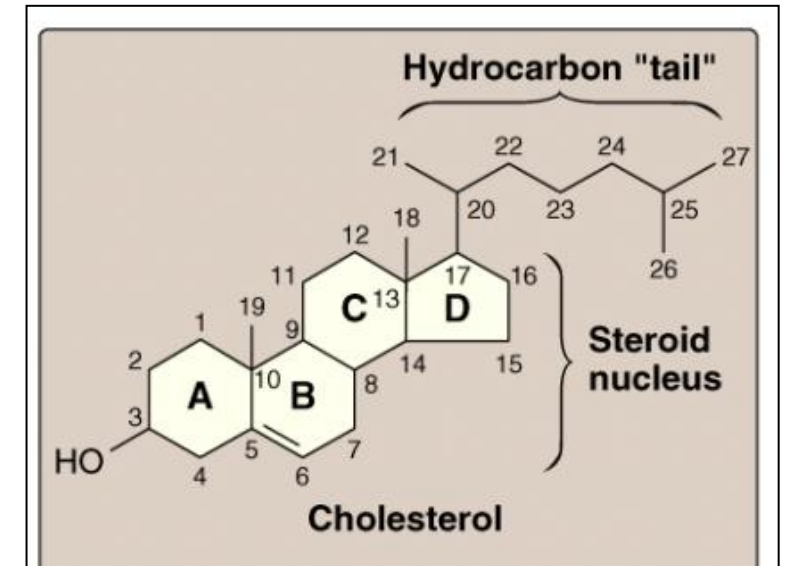
Farnesyl pyrophosphate: **Biological Functions**

- Farnesyl-groups: Needed in specific proteins.
- Farnesyl-PP (FPP; 15-C intermediate) branches out for synthesis of:
 - **Coenzyme Q** (CoQ) of the ETC
 - **Dolichol-PP** used for N-glycosylation of proteins
- **Patients on statins may have reduced CoQ and dolichol-PP synthesis (muscle pain)**



Smith-Lemli-Optiz Syndrome (SLOS)

- Deficiency of 7-Dehydrocholesterol reductase
 - Loss of function mutations
- **7-Dehydrocholesterol reductase** catalyzes the final step in cholesterol synthesis
 - For the correct double bond formation in ring B



<http://www.ambrygen.com/tests/smith-lemli-optiz-syndrome-slos>

SLOS or 7-Dehydrocholesterol Reductase Deficiency

- **Autosomal recessive disorder** (incidence 1 in 20/40,000). High infant mortality rate.
- Heart defects, malformation of limbs, growth retardation, microcephaly (small head) with associated mental disability & ambiguous genitalia in males.
- **Cholesterol administration** helps with growth but not CNS defects due to embryonic microcephaly.



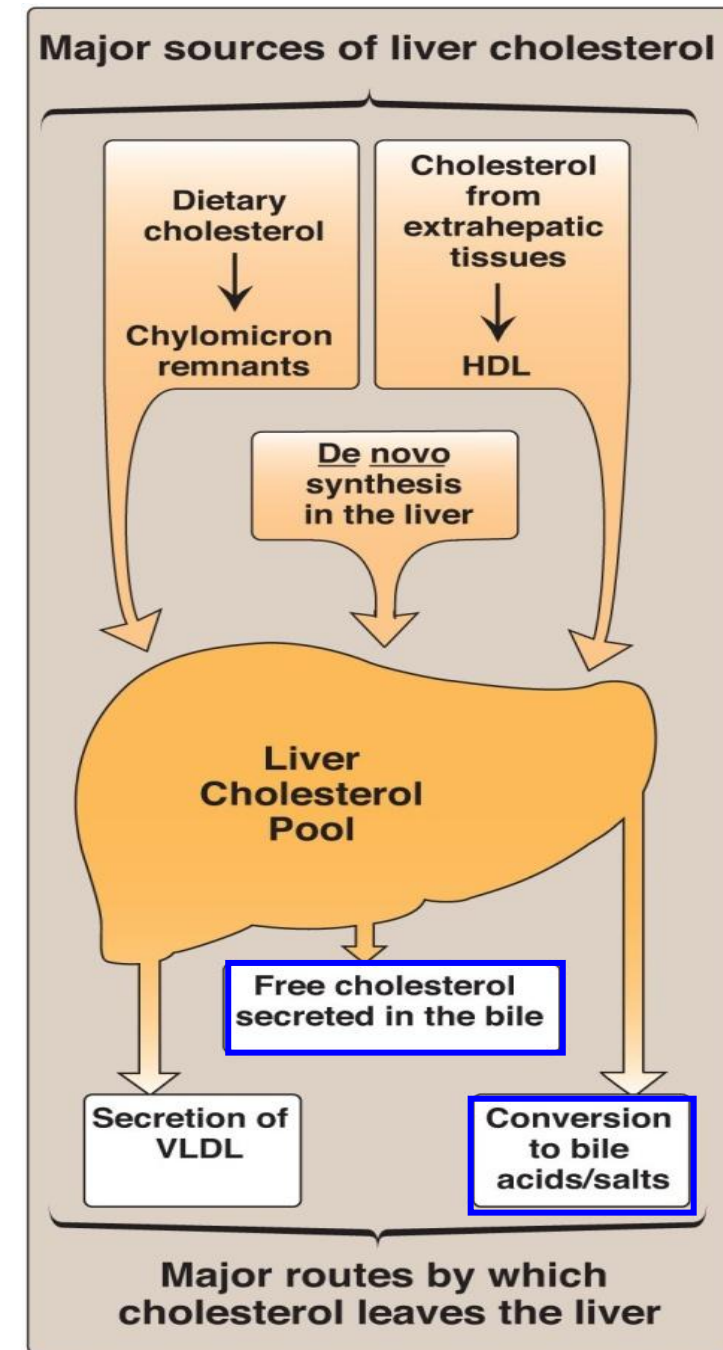
<http://www.bbc.com/news/health-21706761>



<https://www.jci.org/articles/view/29872/figure/1>

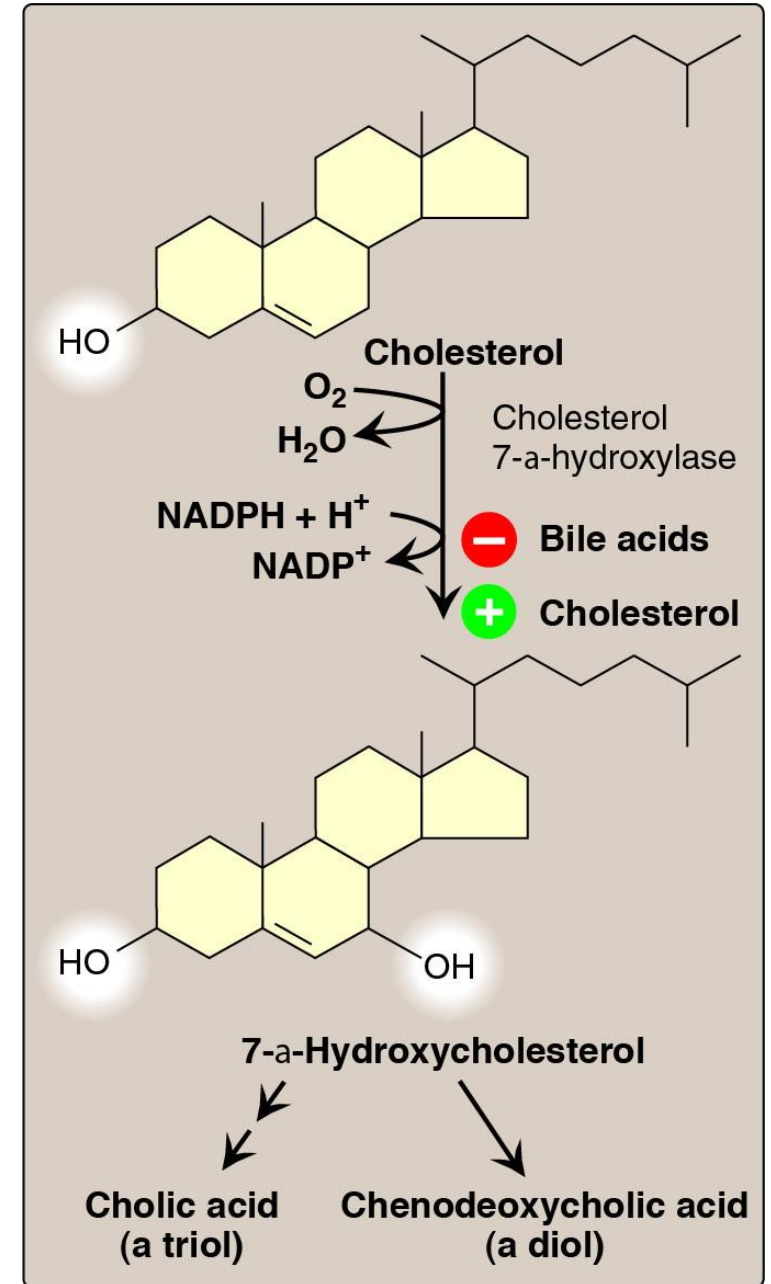
Degradation of Cholesterol

- Cholesterol ring **cannot** be metabolized to CO_2 and H_2O for energy
- **Cholesterol metabolized by:**
 1. **Conversion to bile acids:** secreted in bile and lost in feces
 - Total fecal excretion of **bile salts** balances hepatic synthesis.
 - Major catabolic path in cholesterol metabolism.
 2. **Conversion by intestinal microorganisms** and lost in feces



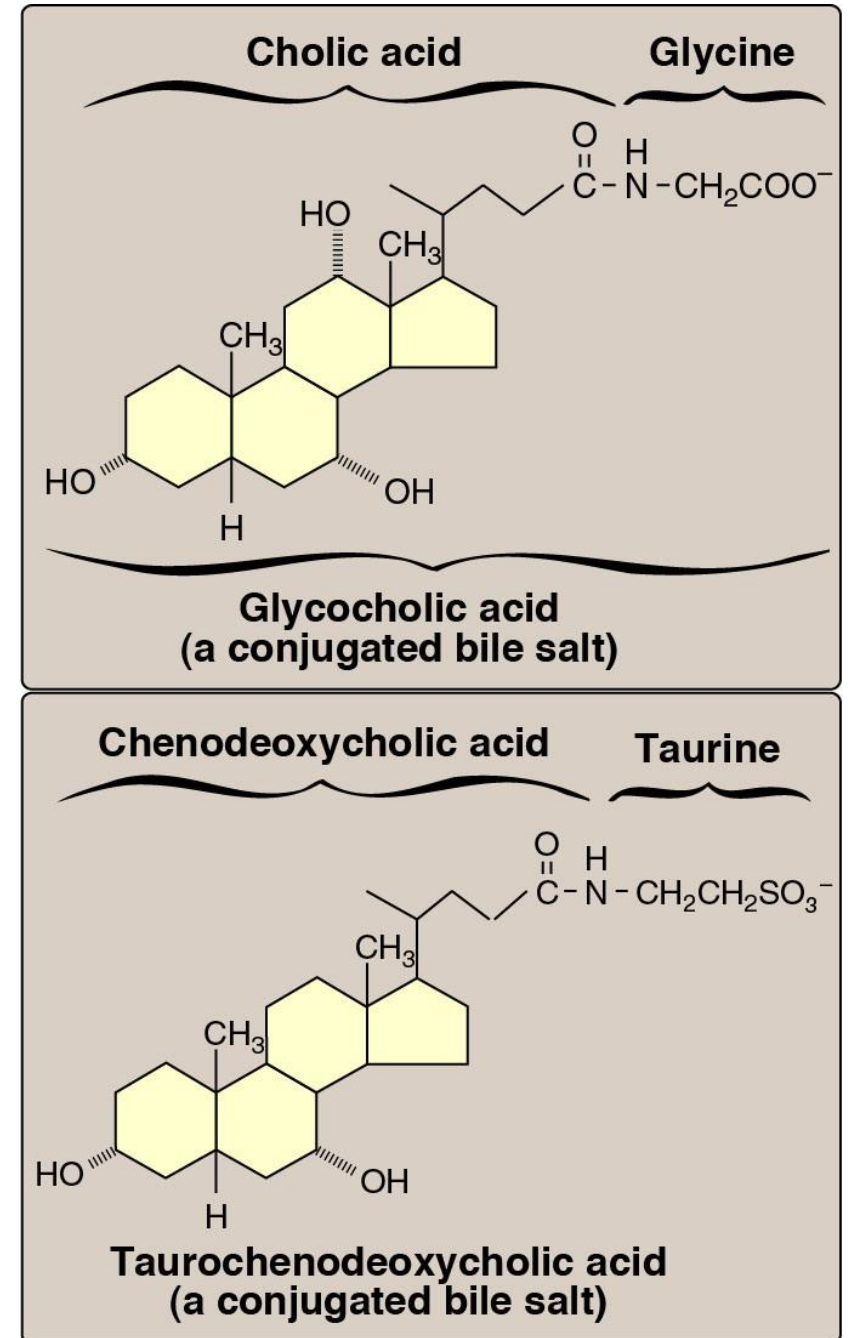
Bile acid synthesis (cholic acid and chenodeoxycholic acid) from cholesterol

- 1st step catalyzed by cholesterol 7 α -hydroxylase (P450)
 - Requires NADPH & O₂
 - Rate limiting step
- Activated by high cholesterol
- Inhibited by high cholic acid (bile acid)
- Products are cholic acid or chenodeoxycholic acid – **Primary bile acids**



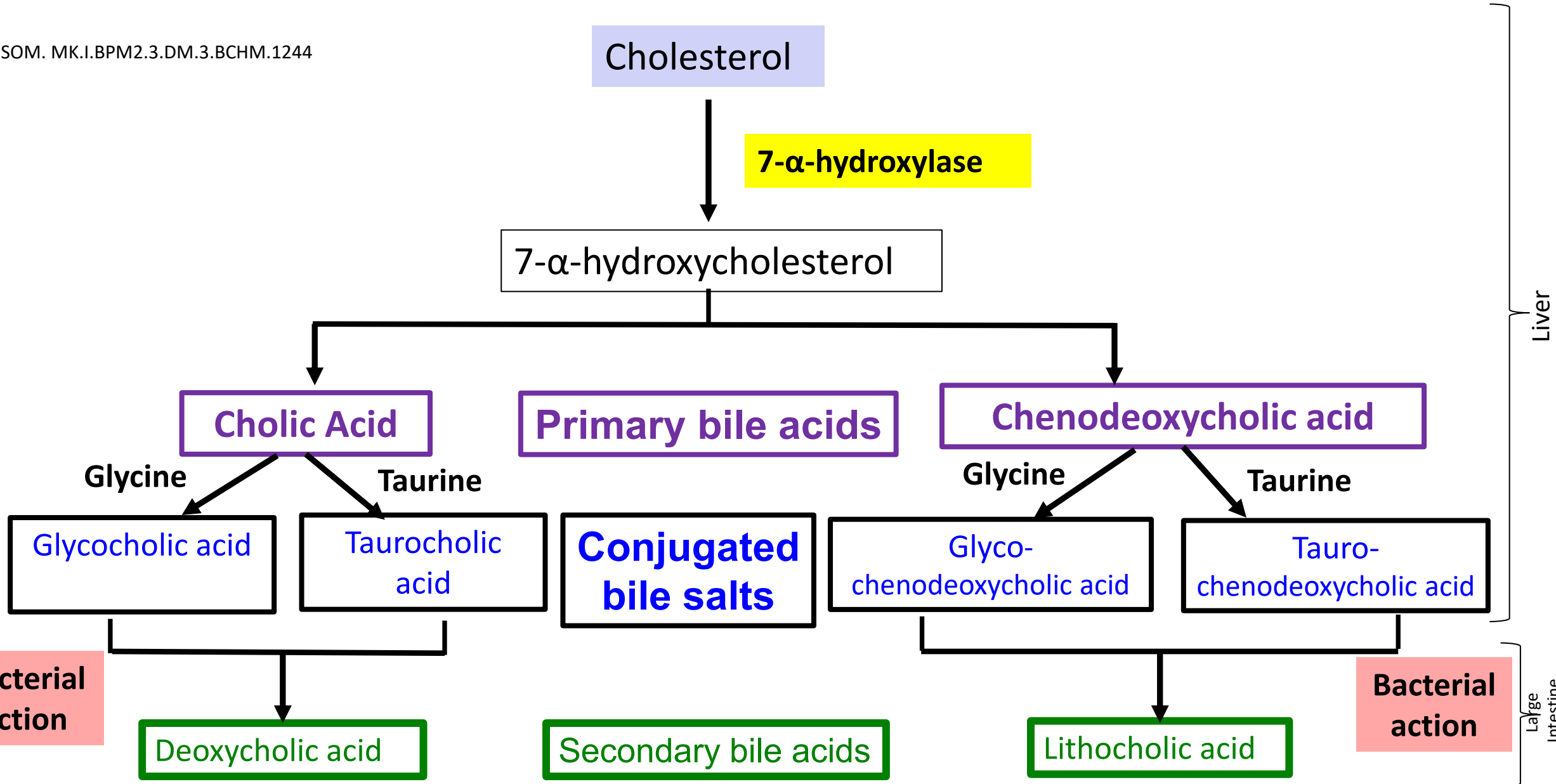
Conjugated Bile Salts

- Conjugated with – glycine or taurine.
 - glycocholic acid and glychenodeoxycholic acid
 - taurocholic acid and taurochenodeoxycholic acid
- Only in hepatocytes.
- Usually present as Na⁺ or K⁺ salts – bile salts
- Conjugated bile salts like glycocholic acid or taurochenodeoxycholic acid solubilize dietary lipids.

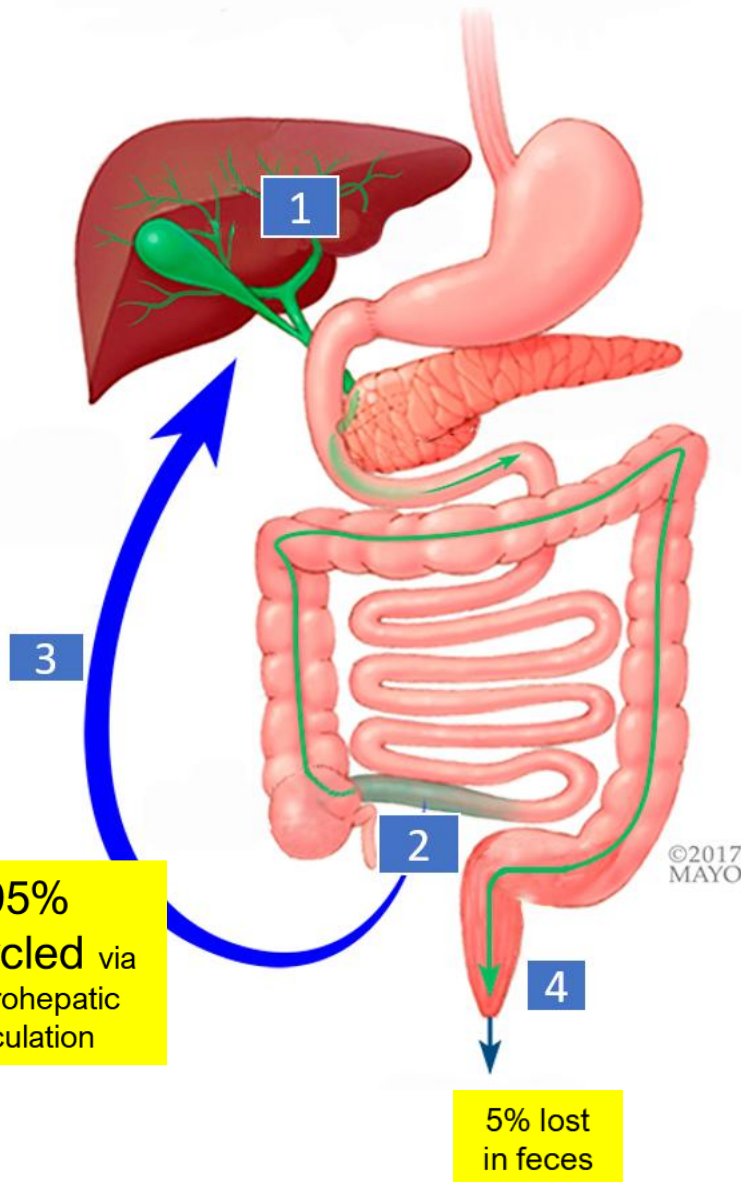


Summary: Metabolism of Bile Acids/Bile Salts

SOM. MK.I.BPM2.3.DM.3.BCHM.1244



Enterohepatic Circulation of Bile Acids/Bile Salts

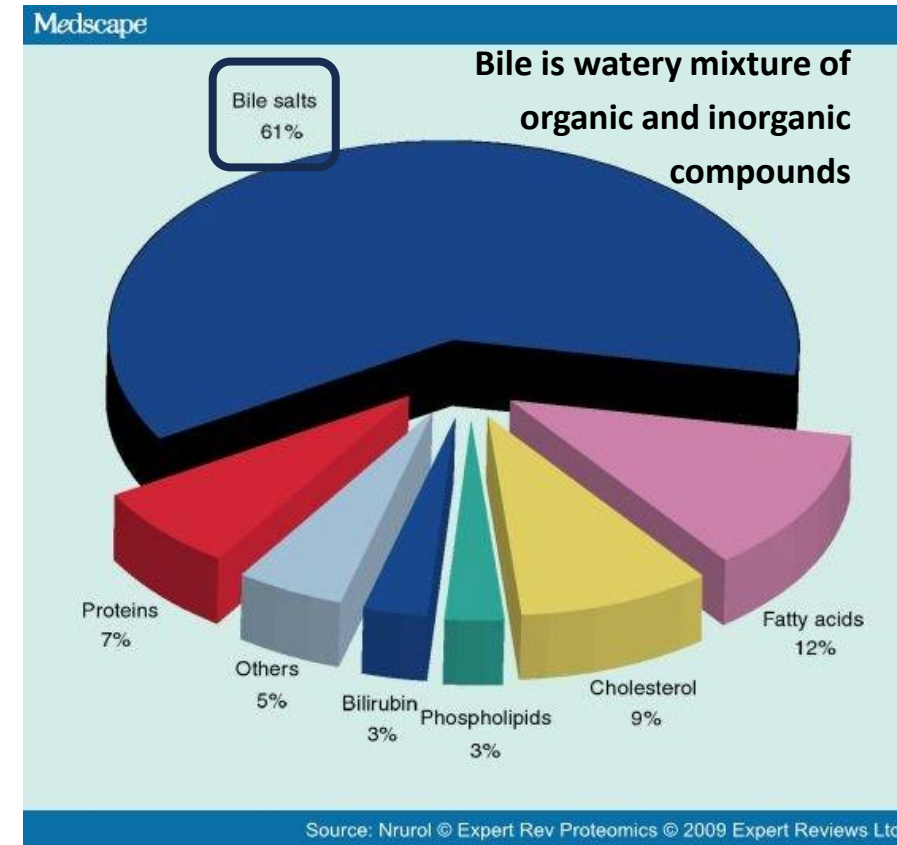


- **Liver conjugates** primary bile acids and **releases Conjugated primary bile salts** into the **bile (1)**
- They assist in **lipid digestion and absorption** in small intestine
- In **terminal ileum** and proximal colon, **most of the primary bile acids are absorbed** (95% of bile salts/acids) [2,3]
- **Enterohepatic circulation:** The cycle of bile salt secretion into bile, passage through the duodenum, uptake into the ileum, and subsequent return to liver
- **Intestinal bacteria** convert cholic acid and chenodeoxycholic acid to generate **secondary bile acids/salts** (deoxycholic acid and lithocholic acid respectively) – Most of which is lost in feces [4]

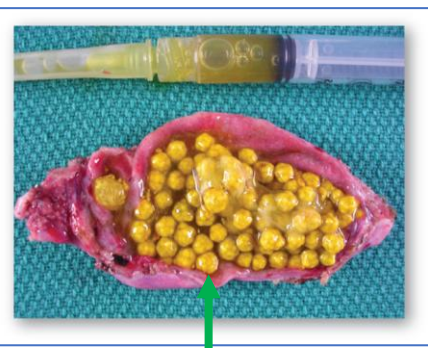
Clinical Significance of Bile Acids/Salts

Composition of Bile

- Bile synthesized in liver and stored in gall bladder
 - **Conjugated bile salts** (glycocholic acid or taurochenodeoxycholic acid) abundant in bile
- Bile enters duodenum through common bile duct
- Functions of Bile acids/salts:
 1. Solubilize cholesterol in bile → **Prevent precipitation of cholesterol in gallbladder (prevent gall stones)**
 2. Facilitate **digestion and absorption dietary triacylglycerols**: Emulsifying agents for action of pancreatic lipase; Form micelles for absorption
 3. Facilitate intestinal absorption of **fat-soluble vitamins**.
 4. Bile acid excretion in feces **eliminates excess cholesterol**.



Gallstones (Cholelithiasis)



Gallbladder with gallstones.

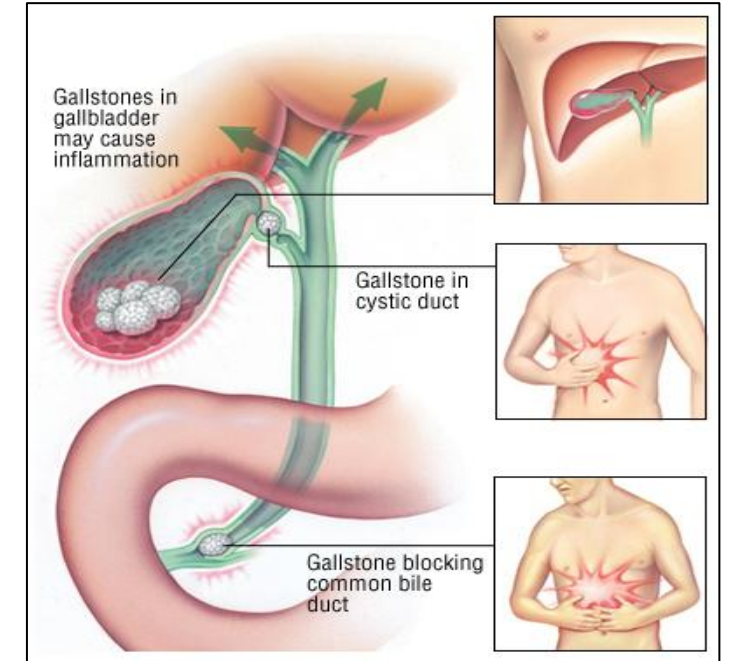
Precipitation of cholesterol (in bile) in gall bladder due to deficiency of bile salts or lecithin

Potential risk factors:

- Decreased bile production (hepatic dysfunction)
- Increased cholesterol content in bile
- Malabsorption of bile acids from the intestine (ileal disease)
- Obstruction of the biliary tract
- Interruption of the enterohepatic circulation of bile acids
- **Diagnosis: CT scan or ultra-sound examination**

Treatment:

- Surgical removal of gall bladder (Laparoscopic cholecystectomy)
- Administration of **chenodeoxycholic acid** (chenodiol)
- Disintegration of stones by shock waves

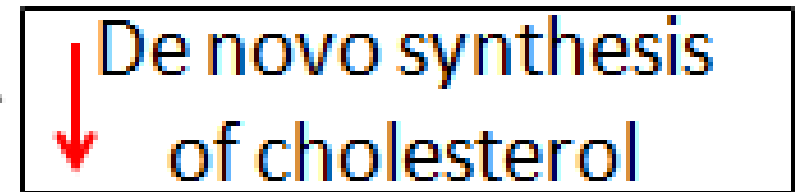
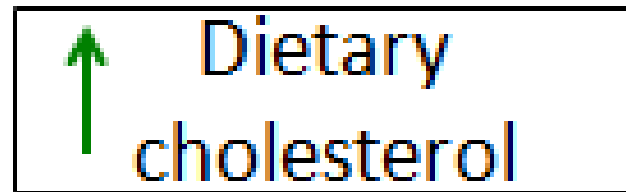


- Gall stones can cause Cholestasis (bile duct obstruction).
- Bile flow blockage leads to lipid malabsorption with steatorrhea (foul-smelling diarrhea).
- Jaundice caused by hyperbilirubinemia. Bile acids also escape into systemic circulation (itching).

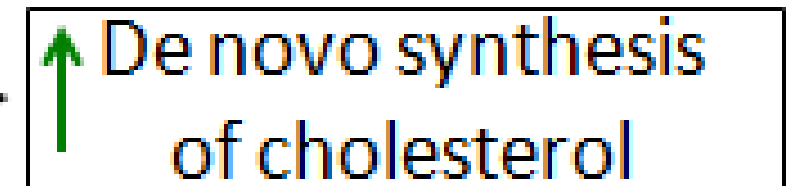
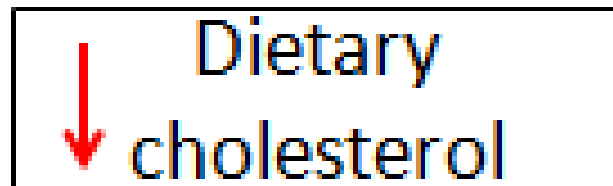
Regulation of Cholesterol Synthesis

- The Rate of *de novo* biosynthesis of cholesterol is related to the amount of dietary cholesterol:

– high intake,
low
de novo synthesis:

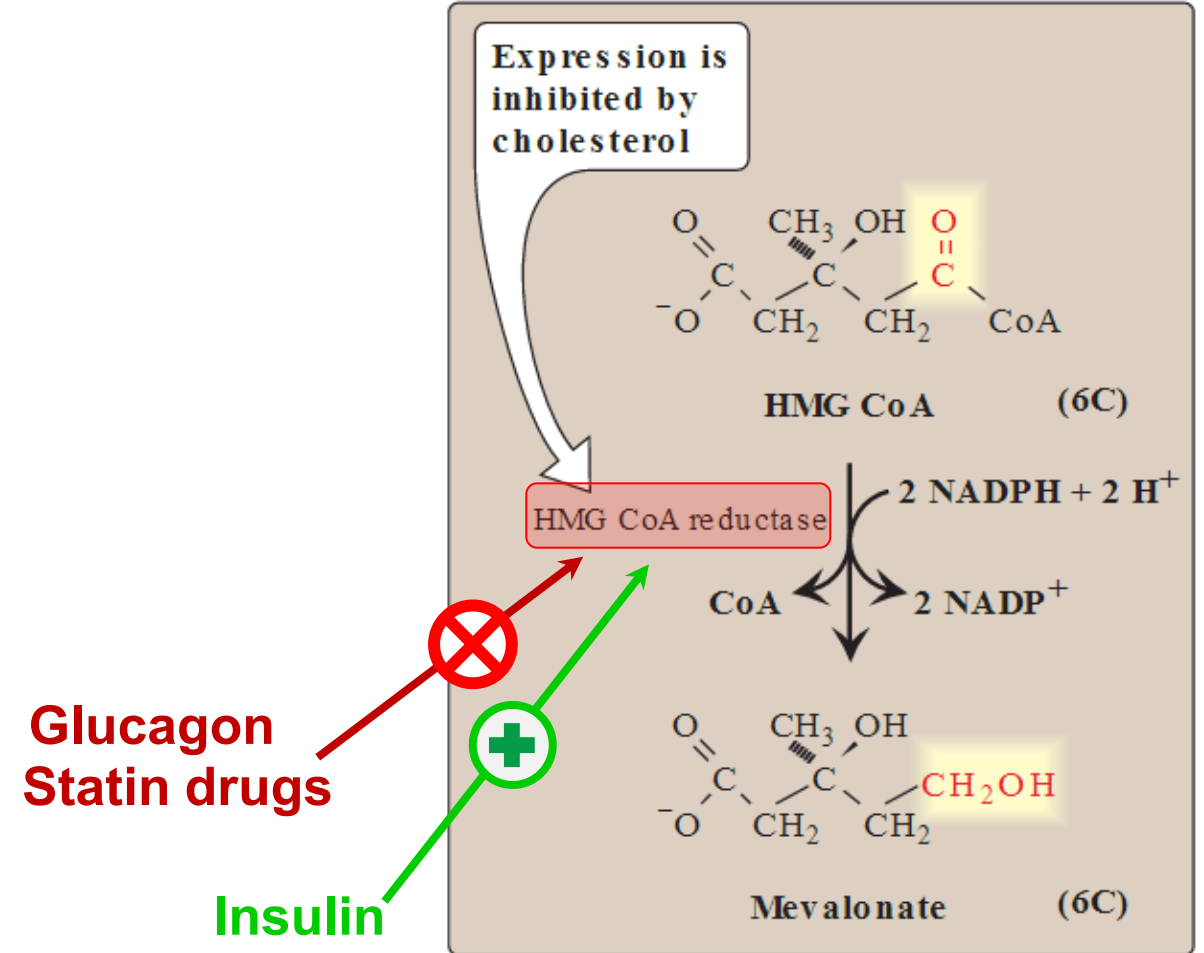


– low intake,
high
de novo synthesis.



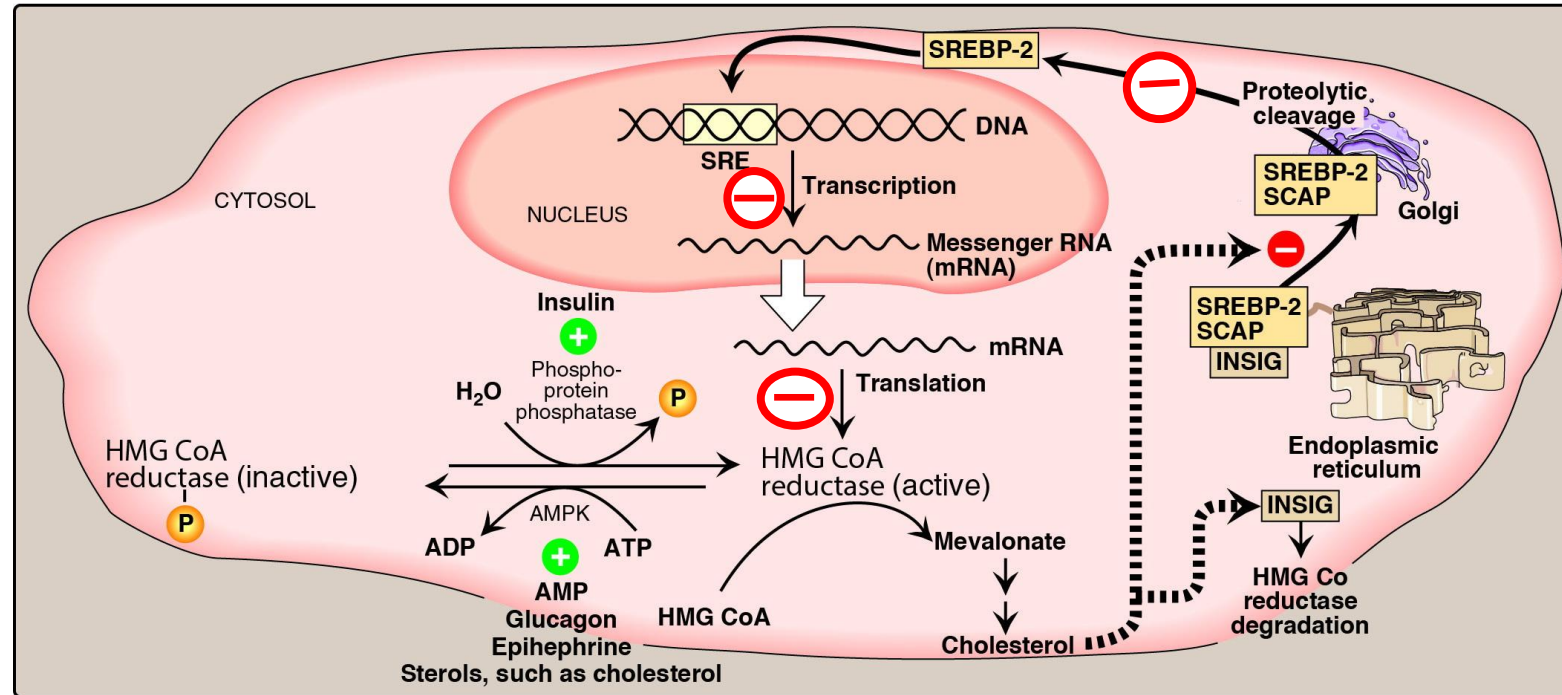
Regulation of cholesterol synthesis: HMG-CoA reductase (Regulated Enzyme)

- Phosphorylation by **AMP-kinase/ Glucagon** inactivates the enzyme
- Cytosolic cholesterol levels regulate:
 - Gene expression (mRNA production)
 - Degradation of HMG-CoA reductase
- Dephosphorylation by **insulin** activates HMG-CoA reductase
- Activated by substrate availability (HMG CoA in cytosol)



Regulation of Gene Expression: High Cytosolic Cholesterol

- **High cytosolic cholesterol** leads to:
 - **Inhibition of transcription** of HMG CoA reductase
 - **Down regulation** of cholesterol synthesis
 - **Early degradation** of enzyme by ubiquitin system in proteasomes



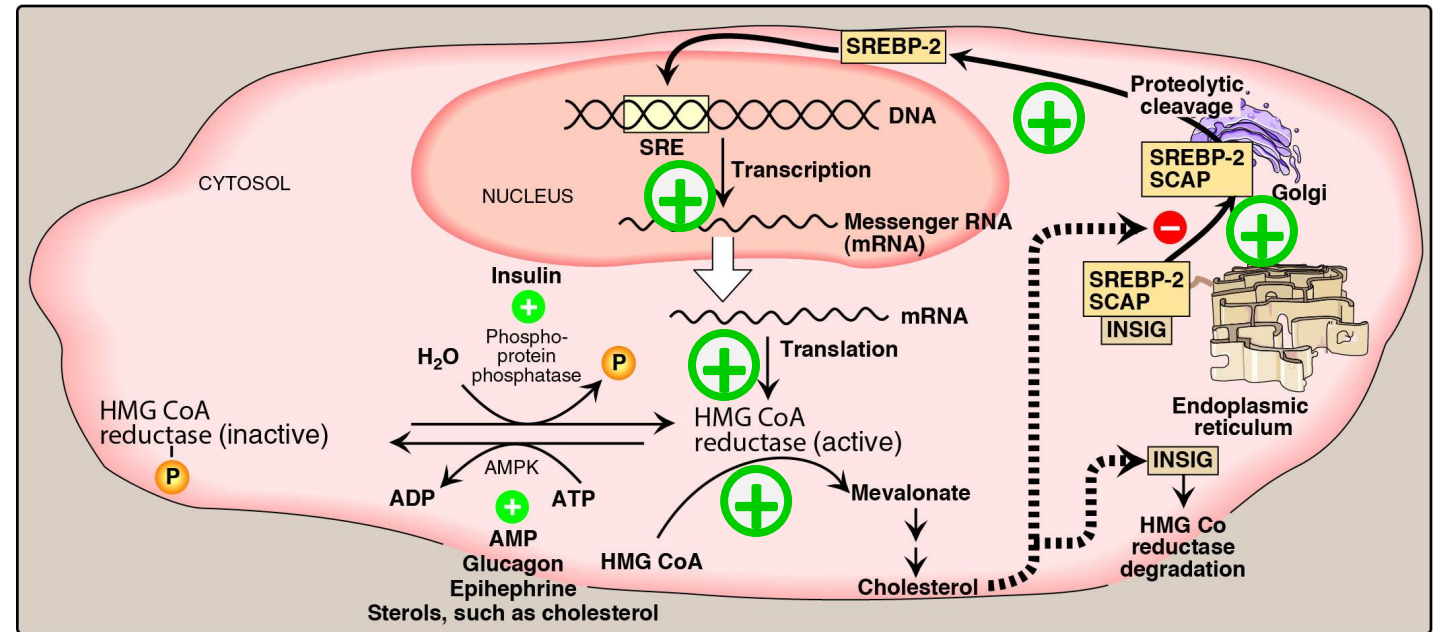
SRE = sterol regulatory element

SREBP = sterol regulatory element binding protein

SCAP = SREBP cleavage-activating protein

Regulation of Gene Expression: Low Cytosolic Cholesterol

- **Low cholesterol** stimulates **release** of regulatory (**SREBP**) protein from ER
- **Increased transcription** of HMG CoA reductase gene and
- **Increase** in amount of HMG CoA reductase in liver



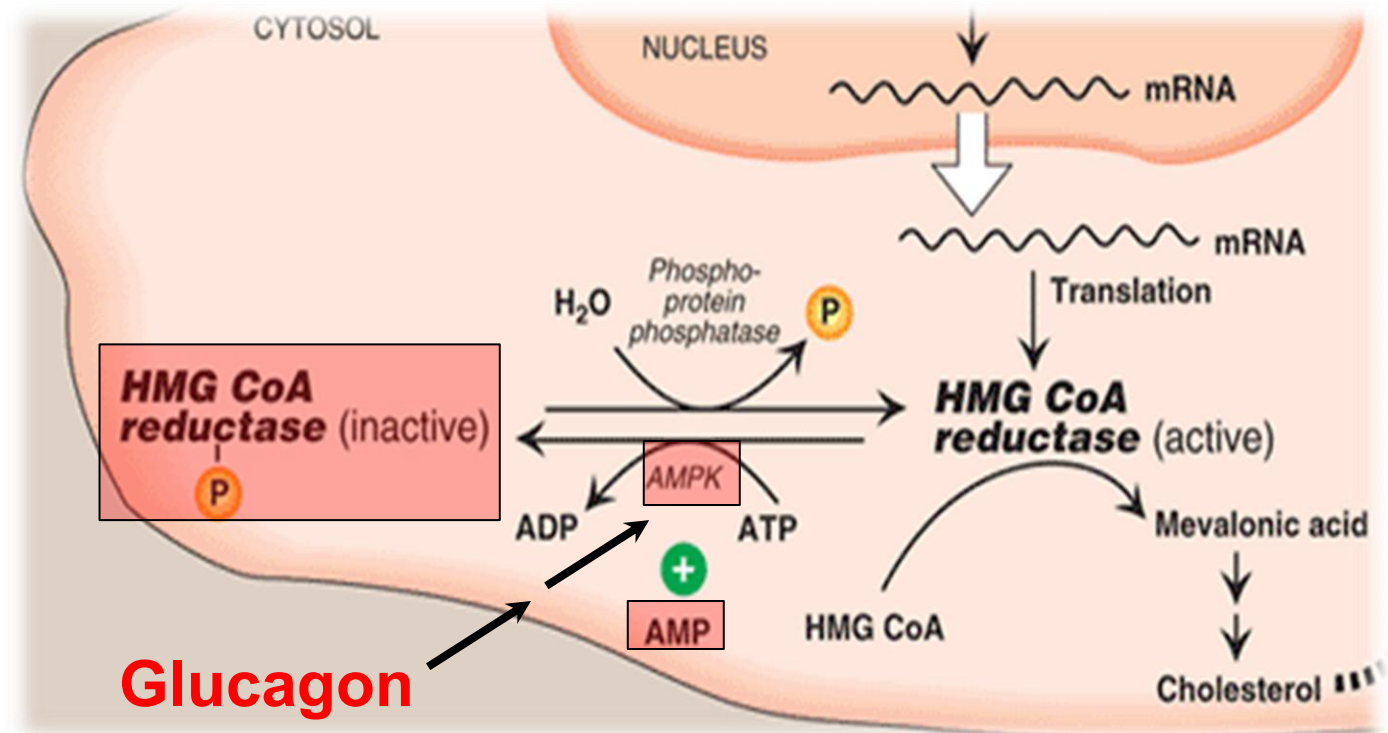
SRE = sterol regulatory element

SREBP = sterol regulatory element binding protein

SCAP = SREBP cleavage-activating protein.

Phosphorylation/Inactivation of HMG-CoA reductase

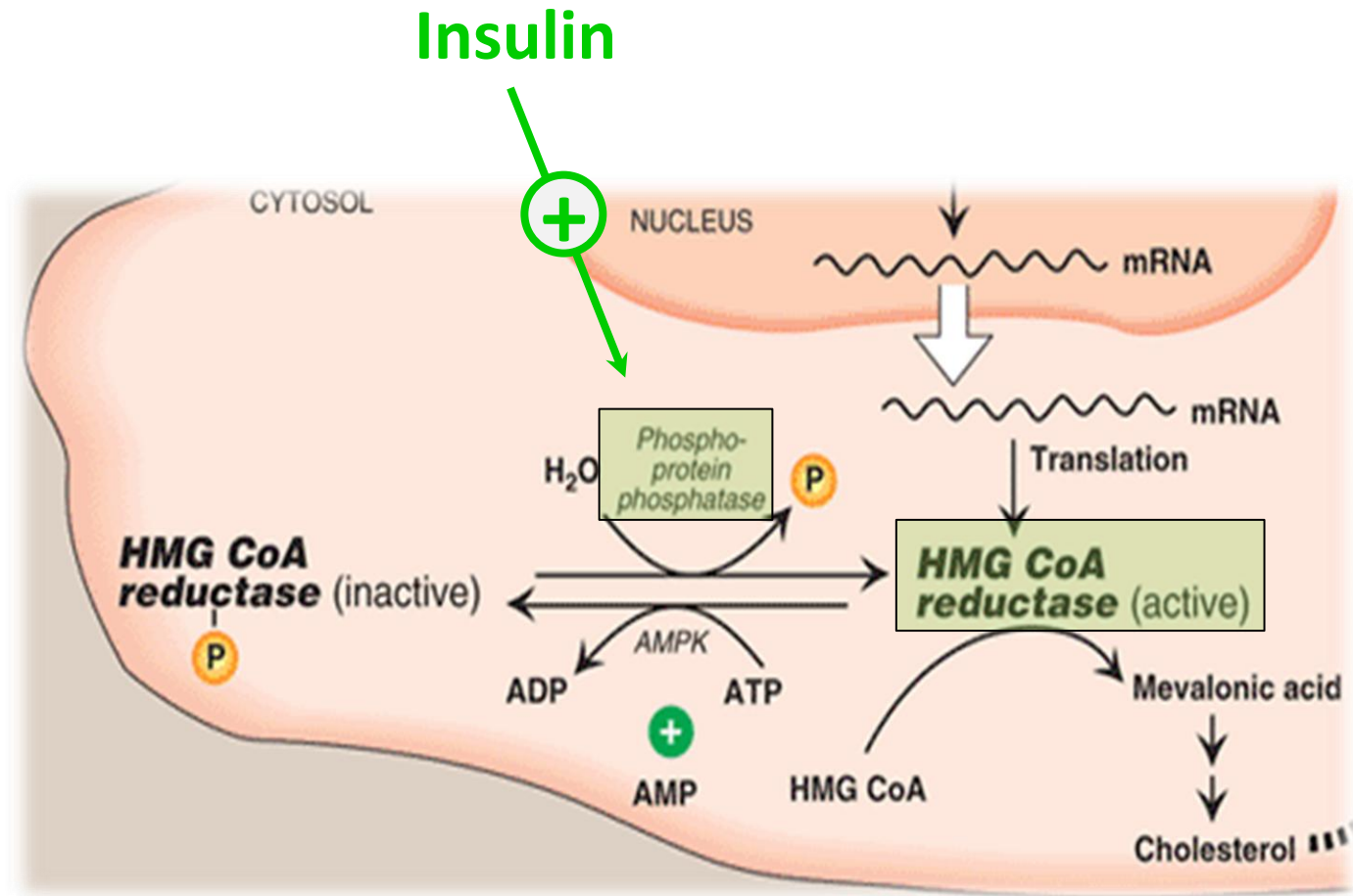
- **AMPK phosphorylates /inactivates** HMG-CoA reductase
- **Glucagon** leads to **phosphorylated inactive** HMG CoA reductase via activation of AMPK
- High intracellular **AMP** (low energy) **stimulate AMPK** (AMP-activated protein kinase)
- Remember: Fatty acid and cholesterol synthesis are inhibited by glucagon (fasting) via AMPK-dependent phosphorylation 😊



AMPK = AMP-activated protein kinase

Regulation by Insulin: **Activation**

- **Short term:** Insulin activates phosphoprotein phosphatase → **Dephosphorylates HMG CoA reductase (active)**
- **Long term:** Insulin favors upregulation of expression of HMG CoA reductase gene

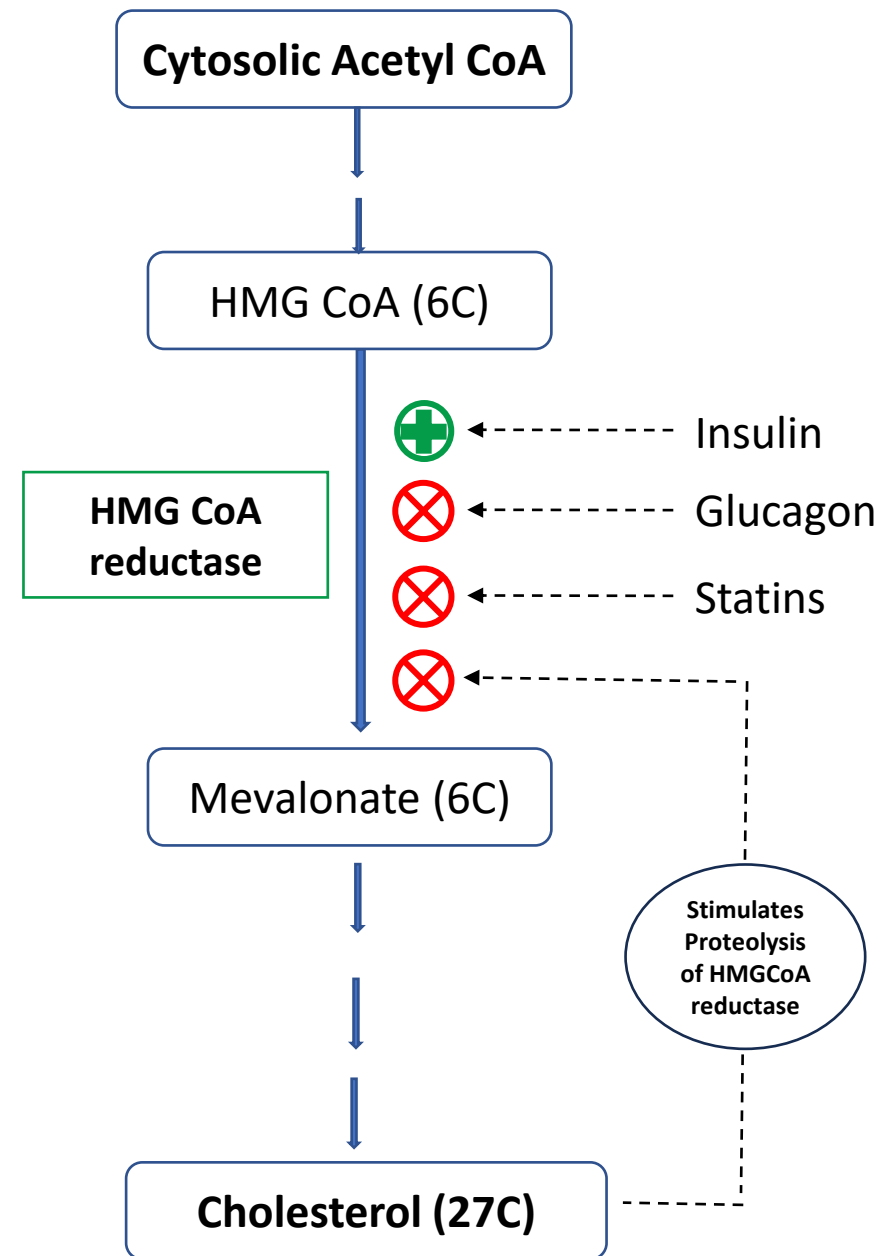


Regulation of *de novo* synthesis of Cholesterol :

Intracellular cholesterol is a key regulator of

➤ HMG-CoA reductase

- Activated by insulin
- Inhibited by glucagon, AMP, AMP-kinase and statins.

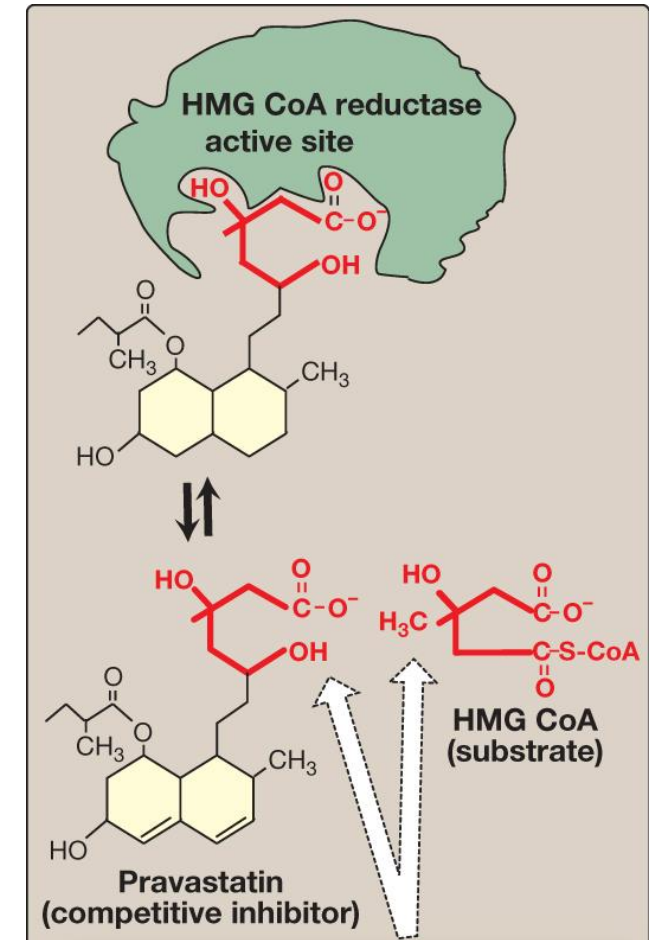


Cholesterol Lowering Therapies

- HMG CoA reductase inhibitors (**Statins**): Competitive reversible inhibition
 - **Cholesterol absorption inhibitors (Ezetimibe)**: Prevents intestinal absorption of cholesterol.
 - **Bile acid sequestrants** (bile acid-binding agents): Enables intestine to get rid of more cholesterol. Prevents enterohepatic circulation of bile salts
 - **ATP-citrate lyase (ACLY) inhibitors**: Reduce formation of cytosolic acetyl CoA
 - **PCSK9 inhibitors**: Increase LDL-receptor recycling
 - **siRNA based therapy**: Prevent expression of PCSK-9 and increase LDL receptor recycling
- 
- Alone or combined

Statin Drugs: Reversible Competitive Inhibitor of HMG CoA Reductase

- HMG-CoA reductase: Key regulated enzyme in cholesterol biosynthesis
- **Statins**: HMG-CoA analogs (substrate analogs)
- **Reversible competitive inhibitors of HMG CoA reductase**
 - Prevent mevalonate/ cholesterol synthesis
 - Used in treatment of hypercholesterolemia



Portions of the statins (shown in red) clearly resemble HMG CoA. However, the bulky hydrophobic group of the inhibitors differ from the CoA moiety of the substrate.

Synthesis of Cholesterol - Overview

- Requires **cytosolic acetyl CoA**, **NADPH** and **ATP (similar to fatty acid synthesis)**
- Rate Limiting Enzyme(RLE): **HMG CoA reductase**
 - Inhibited by **statins** (competitive inhibitor)
 - Regulation: (1) via transcription factor, SREBP-2; (2) accelerated degradation of enzyme when cholesterol levels high; (3) phosphorylation by AMPK/ glucagon → Inactivation of enzyme and (4) Insulin → Dephosphorylation → Activation of enzyme
- Occurs in cytoplasm and Endoplasmic reticulum (ER)
- **Organs: Liver, intestine, adrenal cortex, and gonads**
- Significance of elevated Cholesterol:
 - Elevated plasma cholesterol increases risk of **atherosclerosis**
 - Increased **biliary cholesterol content** increases risk of **gall stones**

***De novo* synthesis of Cholesterol**

- Source of carbon atoms: Acetyl-CoA
- Steps of cholesterol synthesis:
 1. Synthesis of HMG-CoA
 2. **Synthesis of mevalonate (HMG CoA reductase/ uses NADPH)**
 - Inhibited by **statins (used in hypercholesterolemia)**
 3. Formation of a 5-C isopentenyl pyrophosphate
 4. Formation of farnesyl pyrophosphate
 5. Formation of squalene (C30).
 6. Cyclization of squalene to lanosterol followed by cholesterol (C27) formation

Bile acids/salts

- Cholesterol eliminated from body:
 - Conversion to bile salts or
 - Secretion into **bile**
- Bile acid synthesis: **Cholesterol-7- α -hydroxylase (rate limiting enzyme)**
- Before bile acids leave the liver, they are conjugated with glycine and taurine
- Primary Bile salts efficiently reabsorbed (>95%) and return to liver by **enterohepatic circulation**.
- Intestinal **bacteria** modify primary bile salts producing **secondary bile salts**.
- **Most of Secondary bile salts are lost in feces.**
- Enterohepatic circulation of bile salts is reduced by **bile acid sequestrants (lower serum cholesterol)**
- Presence of excess cholesterol in bile increases risk of cholesterol gallstones (cholelithiasis)

Thank you!

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